1 I

1.1 (1)

The linear mass density of the rod is

$$\lambda = \frac{M}{2a}.$$

Thus, the moment of inertia about the pivot is

$$I_0 = \int_0^{2a} \lambda r^2 \, dr = \frac{4}{3} M a^2.$$

1.2 (2)

The equation of rotational motion is

$$I_0 \ddot{\phi} = -Mga \sin\phi \approx -Mga\phi.$$

Therefore,

$$\ddot{\phi} = -\frac{Mga}{I_0}\phi,$$

which gives the angular frequency

$$\boxed{\omega = \sqrt{\frac{Mga}{I_0}}}.$$

1.3 (3)

From the angular impulse–momentum theorem,

$$Pa = I_0 \dot{\phi}_i,$$

where $\dot{\phi}_i$ is the angular velocity of the rod immediately after the impulse.

From energy conservation during the subsequent motion,

$$Mga(1 - \cos\phi_0) = \frac{1}{2} I_0 \dot{\phi}_i^2,$$
$$\frac{1}{2} Mga\phi_0^2 = \frac{1}{2} I_0 \dot{\phi}_i^2.$$

Thus,

$$\boxed{\phi_0 = P\sqrt{\frac{a}{MgI_0}}}.$$

2 II

2.1 (4)

The kinetic energy of the system is

$$\begin{aligned} T = &\frac{1}{2}m\dot{x}^2 \\ &+ \frac{1}{2}M\left[(\dot{x} + a\dot{\theta}\cos\theta)^2 + (a\dot{\theta}\sin\theta)^2\right] \\ &+ \frac{1}{2}I\dot{\theta}^2. \end{aligned}$$

The potential energy is

$$U = -Mga\cos\theta.$$

2.2 (5)

For small oscillations, keeping terms up to second order gives

$$\begin{aligned} T \approx &\frac{1}{2}m\dot{x}^2 \\ &+ \frac{1}{2}M\left[\dot{x}^2 + a^2\dot{\theta}^2 + 2a\dot{x}\dot{\theta}\right] \\ &+ \frac{1}{2}I\dot{\theta}^2, \end{aligned}$$

and

$$U \approx -Mga\left(1 - \frac{\theta^2}{2}\right).$$

Neglecting the constant term, the Lagrangian becomes

$$\boxed{L = \frac{m+M}{2}\dot{x}^2 + \frac{Ma^2 + I}{2}\dot{\theta}^2 + Ma\dot{x}\dot{\theta} - \frac{Mga}{2}\theta^2}.$$

The omitted constant Mga does not affect the equations of motion.

2.3 (6)

Applying the Euler–Lagrange equation to x, we obtain

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = 0,$$

which gives

$$\boxed{(m+M)\ddot{x} + Ma\ddot{\theta} = 0}.$$

Similarly, applying it to θ,

$$\frac{d}{dt}\frac{\partial L}{\partial \dot{\theta}} - \frac{\partial L}{\partial \theta} = 0,$$

we obtain

$$\boxed{(Ma^2 + I)\ddot{\theta} + Ma\ddot{x} + Mga\theta = 0}.$$

2.4 (7)

In the limit $m \to \infty$, the first equation gives

$$\ddot{x} = -\frac{Ma}{m+M}\ddot{\theta} \longrightarrow 0.$$

Since the platform is initially at rest, it remains approximately stationary. Thus, the system reduces to the fixed-pivot pendulum considered in Part I.

The equation of motion becomes

$$(Ma^2 + I)\ddot{\theta} + Mga\theta = 0.$$

Using the parallel-axis theorem,

$$I_0 = I + Ma^2,$$

we obtain

$$\boxed{\omega = \sqrt{\frac{Mga}{I_0}}}.$$

This agrees with the result in Part I.

2.5 (8)

In the limit $m = 0$, the first equation of motion becomes

$$M\ddot{x} + Ma\ddot{\theta} = 0,$$

or

$$\ddot{x} = -a\ddot{\theta}.$$

Substituting this into the second equation gives

$$(Ma^2 + I)\ddot{\theta} - Ma^2\ddot{\theta} + Mga\theta = 0.$$

Therefore,

$$I\ddot{\theta} + Mga\theta = 0,$$

and the angular frequency is

$$\boxed{\omega = \sqrt{\frac{Mga}{I}}}.$$

Physically, since there is no external horizontal force, the horizontal position of the rod's center of mass remains constant.

For small angles,

$$x' = x + a\sin\theta \approx x + a\theta.$$

Thus,

$$x + a\theta = a\theta_0,$$

where the initial conditions have been used.

The platform moves horizontally to compensate for the motion of the rod relative to the pivot. To the leading order, the rod rotates about its center of mass rather than a fixed pivot.

2.6 (9)

Combining the two equations of motion obtained in (6),

$$\begin{cases} (m+M)\ddot{x} + Ma\ddot{\theta} = 0, \\ (Ma^2 + I)\ddot{\theta} + Ma\ddot{x} + Mga\theta = 0, \end{cases}$$

we can eliminate $\ddot{x}$ and obtain

$$\left(Ma^2 + I - \frac{M^2 a^2}{m+M} \right) \ddot{\theta} + Mga\theta = 0.$$

Therefore, the angular frequency is

$$\boxed{\Omega = \sqrt{\frac{Mga}{Ma^2 + I - \dfrac{M^2 a^2}{m+M}}}}.$$

Using the initial conditions

$$\theta(0) = \theta_0, \qquad \dot{\theta}(0) = 0,$$

we obtain

$$\boxed{\theta(t) = \theta_0 \cos(\Omega t)}.$$

Next, integrating the first equation of motion once,

$$(m+M)\dot{x} + Ma\dot{\theta} = 0,$$

where the initial condition $\dot{x}(0) = 0$ has been used.

Integrating once more gives

$$(m+M)x + Ma\theta = Ma\theta_0,$$

where $x(0) = 0$ and $\theta(0) = \theta_0$ have been used.

Substituting the expression for $\theta(t)$, we obtain

$$\boxed{x(t) = \frac{Ma\theta_0}{m+M} \left[1 - \cos(\Omega t) \right]}.$$